

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

```
# Load Iris dataset
iris = load_iris()

X = iris.data
y = iris.target

print("Feature Names:", iris.feature_names)
print("Target Names:", iris.target_names)
```

```
Feature Names: ['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width (cm)']
Target Names: ['setosa' 'versicolor' 'virginica']
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
```

```
scaler = StandardScaler()

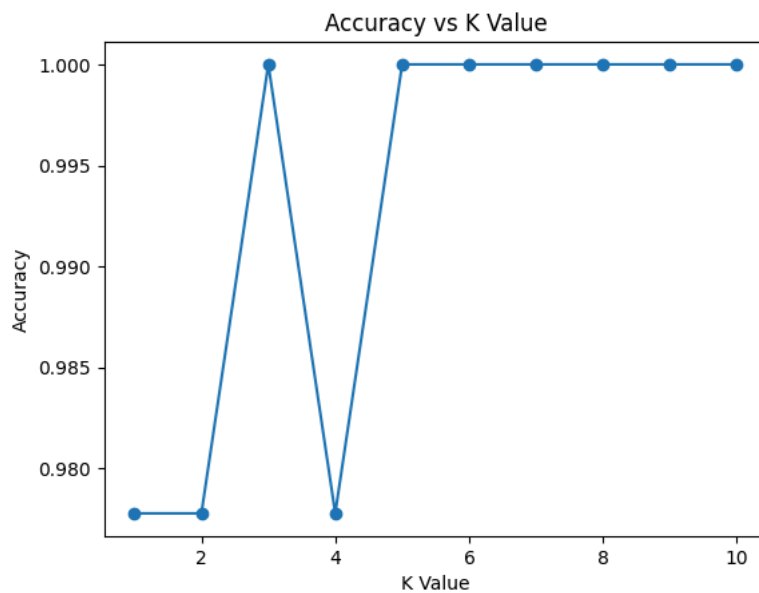
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
accuracies = []

for k in range(1, 11):
    knn = KNeighborsClassifier(n_neighbors=k)
    knn.fit(X_train, y_train)
    y_pred = knn.predict(X_test)
    acc = accuracy_score(y_test, y_pred)
    accuracies.append(acc)
    print(f"K={k}, Accuracy={acc:.2f}")
```

```
K=1, Accuracy=0.98
K=2, Accuracy=0.98
K=3, Accuracy=1.00
K=4, Accuracy=0.98
K=5, Accuracy=1.00
K=6, Accuracy=1.00
K=7, Accuracy=1.00
K=8, Accuracy=1.00
K=9, Accuracy=1.00
K=10, Accuracy=1.00
```

```
plt.figure()
plt.plot(range(1, 11), accuracies, marker='o')
plt.title("Accuracy vs K Value")
plt.xlabel("K Value")
plt.ylabel("Accuracy")
plt.show()
```



```
best_k = accuracies.index(max(accuracies)) + 1
print("Best K:", best_k)

final_model = KNeighborsClassifier(n_neighbors=best_k)
final_model.fit(X_train, y_train)

y_pred = final_model.predict(X_test)
```

Best K: 3

```
print("Final Accuracy:", accuracy_score(y_test, y_pred))
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))
```

Final Accuracy: 1.0

Confusion Matrix:

```
[[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]
```

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	19
1	1.00	1.00	1.00	13
2	1.00	1.00	1.00	13
accuracy			1.00	45
macro avg	1.00	1.00	1.00	45
weighted avg	1.00	1.00	1.00	45

```
X = iris.data[:, :2] # Take first 2 features
y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train, y_train)

# Create mesh
h = .02
x_min, x_max = X_train[:, 0].min() - 1, X_train[:, 0].max() + 1
y_min, y_max = X_train[:, 1].min() - 1, X_train[:, 1].max() + 1
xx, yy = np.meshgrid(np.arange(x_min, x_max, h),
                     np.arange(y_min, y_max, h))

Z = knn.predict(np.c_[xx.ravel(), yy.ravel()])
Z = Z.reshape(xx.shape)
```

```
plt.figure()  
plt.contourf(xx, yy, Z, alpha=0.4)  
plt.scatter(X_train[:, 0], X_train[:, 1], c=y_train)  
plt.title("KNN Decision Boundary")  
plt.show()
```

